SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure C

Case study

Criteria	Understanding of Case ***(2.5)***	Application of Concepts ***(2.5)***	Analysis ***(2)***	Solution Design ***(2)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						

Code of Conduct:

I _______N.Vijetha (192311258)_____________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2.Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3.An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)
- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	**Free**	**Length**	**Class**
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
    - P(Spam), P(Not Spam)
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
    - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

(01) A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them. one night. the shepherd boy sees a real wolf approaching the flock & calls out, "wolf"! The villagers refuse to be fooled again & stayed in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. panci ensures. identify TP, FP, FN, TN.

A \ P	wolf (warning) yes	wolf (warning) NO
wolf (yes)	TP Real wolf came, warning also given	FN No, wolf came, false warning sign.
NO wolf (NO)	FP wolf came, but warning not acted upon (or) wolf came, no warning given	TN No wolf, no warning

(02) Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. it also predicts absence of a cat in 50 of the 70 no cat images. calculate Identify TP, FP, FN, TN and calculate the Precision and Recall and F_1 score

Total images = 100

Actual cat images = 70

A \ P	C	7C	
C	TP 25	FN 20	P 45
7C	FP 5	TN 50	N 55
	P' 30	N' 70	All 100

(i) Precision: $\frac{TP}{TP + FP}$

$\Rightarrow \frac{25}{25+5}$

$\Rightarrow \frac{25}{30}$

$\Rightarrow 0.83 \Rightarrow 83\%$

(ii) Recall: $\frac{TP}{TP+FN}$

$\Rightarrow \frac{25}{25+20}$

$\Rightarrow \frac{25}{45}$

$\Rightarrow 0.55 \Rightarrow 55\%$

(iii) F_1 score: $\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

$\Rightarrow \frac{2 \times 0.83 \times 0.55}{0.83 + 0.55}$

$\Rightarrow \frac{1.66 \times 0.55}{1.38}$

$\Rightarrow \frac{0.913}{1.38}$

$\Rightarrow 0.66$

$\Rightarrow 66\%$

(Q3) An email service provider wants to classify emails as spam or Not spam using features:

- contains "offer" (yes/NO)
- contains "free" (yes/NO)
- Email length (short / long)

sample Dataset

offer	free	Length	class
yes	yes	short	spam
yes	NO	Long	Not spam
NO	yes	short	spam
No	NO	Long	Not spam
yes	yes	Long	spam
NO	NO	short	Not spam

(i) calculate prior probabilities:

- P(spam), P(Not spam)

(ii) compute conditional probabilities for each feature.

(iii) using Naive Bayes, classify:

- (offer = yes, free = NO, Length = short)

(i) Prior probabilities:

$\Rightarrow$ Total emails = 6

- spam = 3
- No spam = 3

$\Rightarrow P(spam) = \frac{3}{6}$

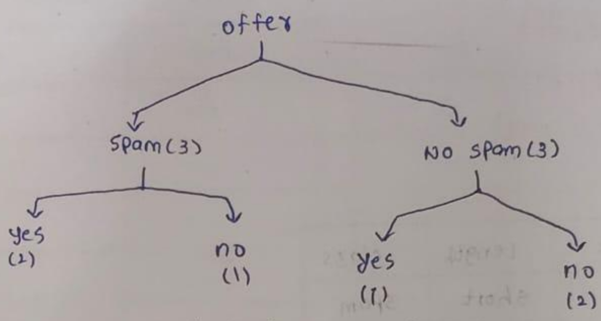
$= 0.5$

$\Rightarrow P(Not\ spam) = \frac{3}{6}$

$= 0.5$

(ii) conditional probabilities:

* offer:



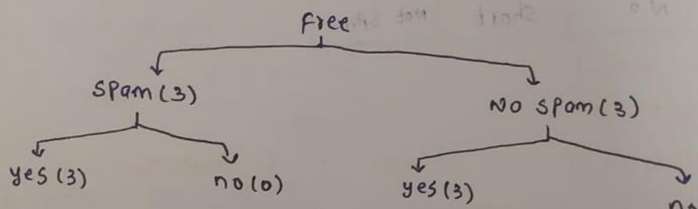
P(A|B) = P((offer = yes)/ spam) $\Rightarrow 2/3$

P(offer=no/ spam) $= 1/3$

P(offer=yes/ nospam) $= 1/3$

P (offer = no/ nospam) $= 2/3$

* Free:



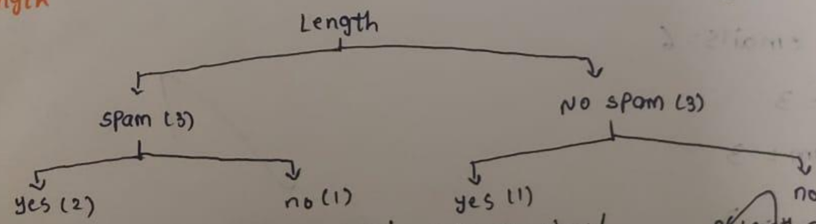
P(A|B) = P(offer=yes/ spam) $= 3/3 = 1$

P(offer=no/ spam) $= 0/3 = 0$

P(offer = yes nospam) $= 3/3 = 1$

P(offer=no/ nospam) $= 0/3 = 0$

* Length



P(A|B) = P(length short|spam) $= 2/3$

P(length = long/ spam) $= 1/3$

P(length = short/ no spam) $= 1/3$

P(length = long/ nospam) $= 2/3$

Feature	value	P(value / spam)	P(value/ Not spam)
offer	Yes	2/3	1/3
	No	1/3	2/3
Free	Yes	3/3	0/3
	No	0/3	3/3
Length	short	2/3	1/3
	Long	1/3	2/3

(ii) classify the new Email:

New Email:

$$X = (\text{offer} = \text{yes},\ \text{free} = \text{No},\ \text{Length} = \text{short})$$

using Naive Bayes:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

P(A) given mail = spam

$$\Rightarrow P(\text{spam}|X) \propto P(\text{spam}) \times P(\text{yes}|\text{spam}) \times P(\text{No}|\text{spam}) \times P(\text{short}|\text{spam})$$

$$\Rightarrow \frac{3}{6} \times \frac{2}{3} \times \frac{0}{3} \times \frac{2}{3}$$

$$\Rightarrow 0$$

P(A) given mail = Not spam

$$\Rightarrow P(\text{Not spam}|X) \propto P(\text{Not spam}) \times P(\text{yes}|\text{not spam}) \times P(\text{no}|\text{not spam}) \times P(\text{short}|\text{Not spam})$$

$$\Rightarrow \frac{3}{6} \times \frac{1}{3} \times \frac{3}{3} \times \frac{1}{3}$$

$$= \frac{1}{18} \Rightarrow 0.0556$$

∵ New Email is classified as Not spam

∴ $P(\text{Not spam}|X) > P(\text{spam}|X)$

(iv) comparing results using weka:

Dataset	(1)	BayesNet	NaiveBayes
Iris	(100)	93.20	95.53
Supermarket	(100)	63.71	63.71
		(v// *)	(0/2/0)

key:

(i) Bayes. BayesNet

(ii) Bayes. Naive Bayes

* ROC curve:

X: false positive Rate (Num)   Y: True positive Rate (Num)

color: Threshold (Num)